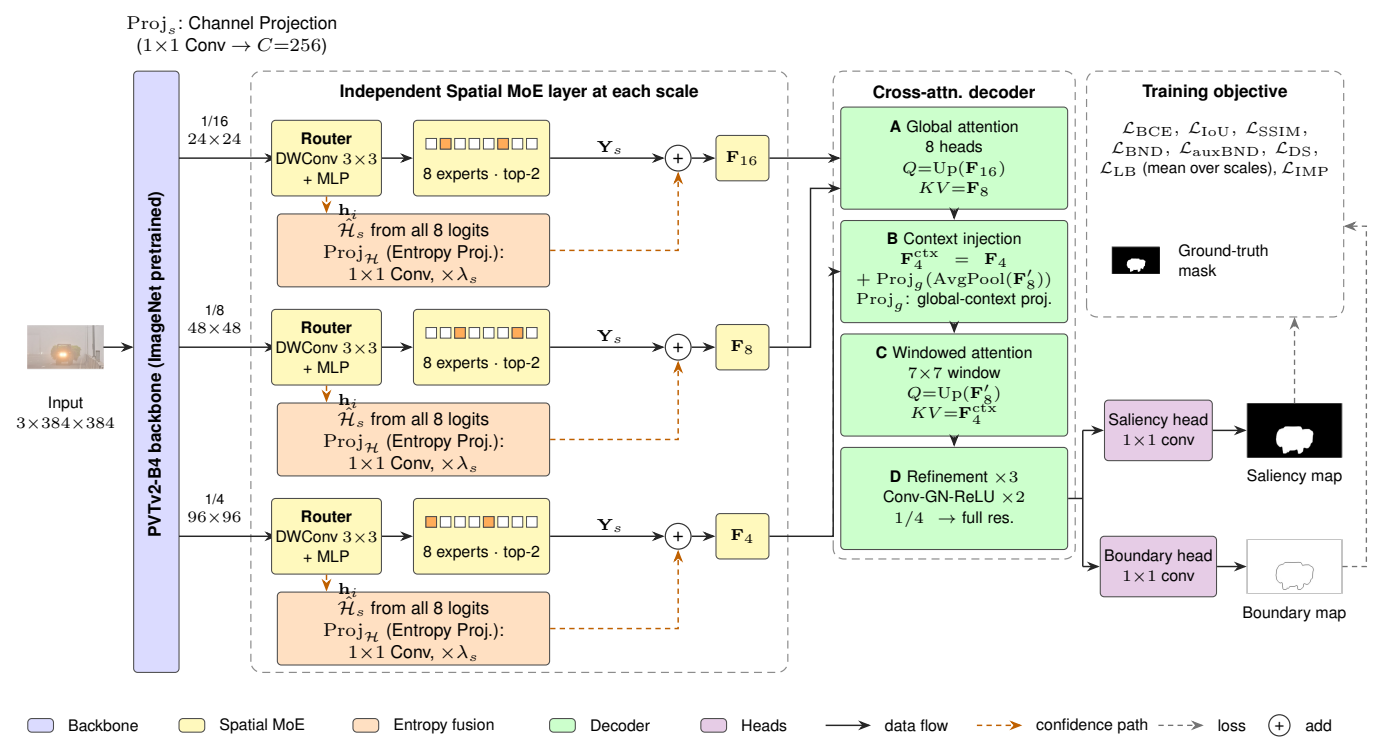

Proj_s: Channel Projection
(1×1 Conv $\rightarrow C{=}256$)
Input
$3\times384\times384$
PVTv2-B4 backbone (ImageNet pretrained)
1/16
24×24
1/8
48×48
1/4
96×96
Independent Spatial MoE layer at each scale
Router
DWConv 3×3
+ MLP
8 experts · top-2
$\mathbf{Y}_s$
$\mathbf{h}_i$
$\hat{\mathcal{H}}_s$ from all 8 logits
$\mathrm{Proj}_{\mathcal{H}}$ (Entropy Proj.):
1×1 Conv, $\times\lambda_s$
$\mathbf{F}_{16}$
$\mathbf{F}_8$
$\mathbf{F}_4$
Cross-attn. decoder
A Global attention
8 heads
$Q{=}\mathrm{Up}(\mathbf{F}_{16})$
$KV{=}\mathbf{F}_8$
B Context injection
$\mathbf{F}_4^{\mathrm{ctx}} = \mathbf{F}_4$
$+\,\mathrm{Proj}_g(\mathrm{AvgPool}(\mathbf{F}'_8))$
Proj_g: global-context proj.
C Windowed attention
7×7 window
$Q{=}\mathrm{Up}(\mathbf{F}'_8)$
$KV{=}\mathbf{F}_4^{\mathrm{ctx}}$
D Refinement $\times3$
Conv-GN-ReLU $\times2$
$1/4 \rightarrow$ full res.
Training objective
$\mathcal{L}_{\mathrm{BCE}}$, $\mathcal{L}_{\mathrm{IoU}}$, $\mathcal{L}_{\mathrm{SSIM}}$,
$\mathcal{L}_{\mathrm{BND}}$, $\mathcal{L}_{\mathrm{auxBND}}$, $\mathcal{L}_{\mathrm{DS}}$,
$\mathcal{L}_{\mathrm{LB}}$ (mean over scales), $\mathcal{L}_{\mathrm{IMP}}$
Ground-truth mask
Saliency head
1×1 conv
Saliency map
Boundary head
1×1 conv
Boundary map
Backbone
Spatial MoE
Entropy fusion
Decoder
Heads
data flow
confidence path
loss
add